

EXPERIMENT - 4

Aim:-

Using the Employee schema, demonstrate:

- INNER JOIN
- LEFT JOIN
- Self Join
- 3-way Join
- Correlated Subquery
- EXISTS
- Simulated INTERSECT
- Simulated EXCEPT
- EXPLAIN execution plans

Source Code:-

INNER JOIN

```
SELECT
    e.emp_name,
    d.dept_name
FROM Employee e
INNER JOIN Department d
    ON e.dept_id = d.dept_id;
```

LEFT JOIN

```
SELECT
    d.dept_name,
    e.emp_name
FROM Department d
LEFT JOIN Employee e
    ON d.dept_id = e.dept_id;
```

SELF JOIN

```
SELECT
    e1.emp_name AS employee_1,
    e2.emp_name AS employee_2,
    e1.dept_id
FROM Employee e1
JOIN Employee e2
    ON e1.dept_id = e2.dept_id
    AND e1.emp_id < e2.emp_id;
```

3-WAY JOIN

```
SELECT
    e.emp_name,
    d.dept_name,
    p.project_name,
    e.salary
FROM Employee e
JOIN Department d
    ON e.dept_id = d.dept_id
JOIN Project p
    ON e.project_id = p.project_id;
```

CORRELATED SUBQUERY

```
SELECT
    e.emp_name,
    e.salary,
    e.dept_id
FROM Employee e
WHERE e.salary >
(
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.dept_id = e.dept_id
);
```

EXISTS

```
SELECT
    d.dept_id,
    d.dept_name
FROM Department d
WHERE EXISTS
(
    SELECT 1
    FROM Employee e
    WHERE e.dept_id = d.dept_id
        AND e.salary > 90000
);
```

SIMULATED INTERSECT

```
SELECT e.emp_id, e.emp_name
FROM Employee e
WHERE e.dept_id = 2

AND e.emp_id IN
(
    SELECT emp_id
    FROM Employee
    WHERE salary > 70000
);
```

SIMULATED EXCEPT

```
SELECT emp_id, emp_name
FROM Employee
WHERE dept_id = 2

AND emp_id NOT IN
(
    SELECT emp_id
    FROM Employee
    WHERE salary > 70000
);
```

EXPLAIN - INNER JOIN

EXPLAIN

SELECT

 e.emp_name,

 d.dept_name

FROM Employee e

INNER JOIN Department d

 ON e.dept_id = d.dept_id;

EXPLAIN - LEFT JOIN

EXPLAIN

SELECT

 d.dept_name,

 e.emp_name

FROM Department d

LEFT JOIN Employee e

 ON d.dept_id = e.dept_id;

Output:-

INNER JOIN

emp_name	dept_name
Aarav Sharma	Computer Science Engineering
Rohan Verma	Computer Science Engineering
Isha Patel	Computer Science Engineering
Arjun Das	Computer Science Engineering
Nikhil Jain	Computer Science Engineering
Tanya Roy	Computer Science Engineering
Dev Thakur	Computer Science Engineering
Ananya Singh	Sciences
Rahul Joshi	Sciences
Vikram Rao	Sciences
Simran Kaur	Sciences
Yash Agarwal	Sciences
Nandini Iyer	Sciences
Priya Mehta	Law
Neha Kapoor	Law

LEFT JOIN

dept_name	emp_name
Computer Science Engineering	Aarav Sharma
Computer Science Engineering	Rohan Verma
Computer Science Engineering	Isha Patel
Computer Science Engineering	Arjun Das
Computer Science Engineering	Nikhil Jain
Computer Science Engineering	Tanya Roy
Computer Science Engineering	Dev Thakur
Sciences	Ananya Singh
Sciences	Rahul Joshi
Sciences	Vikram Rao
Sciences	Simran Kaur
Sciences	Yash Agarwal
Sciences	Nandini Iyer
Law	Priya Mehta
Law	Neha Kapoor

SELF JOIN

employee_1	employee_2	dept_id
Aarav Sharma	Rohan Verma	1
Aarav Sharma	Isha Patel	1
Aarav Sharma	Arjun Das	1
Aarav Sharma	Nikhil Jain	1
Aarav Sharma	Tanya Roy	1
Aarav Sharma	Dev Thakur	1
Ananya Singh	Rahul Joshi	2
Ananya Singh	Vikram Rao	2
Ananya Singh	Simran Kaur	2
Ananya Singh	Yash Agarwal	2
Ananya Singh	Nandini Iyer	2
Rohan Verma	Isha Patel	1
Rohan Verma	Arjun Das	1
Rohan Verma	Nikhil Jain	1
Rohan Verma	Tanya Roy	1

3-WAY JOIN

emp_name	dept_name	project_name	salary
Aarav Sharma	Computer Science Engineering	Campus Management System	65000.00
Rohan Verma	Computer Science Engineering	Cyber Security Platform	58000.00
Isha Patel	Computer Science Engineering	Campus Management System	61000.00
Arjun Das	Computer Science Engineering	Cyber Security Platform	63000.00
Nikhil Jain	Computer Science Engineering	Campus Management System	66000.00
Tanya Roy	Computer Science Engineering	Cyber Security Platform	62000.00
Dev Thakur	Computer Science Engineering	Campus Management System	64000.00
Ananya Singh	Sciences	AI Chatbot	72000.00
Rahul Joshi	Sciences	Computer Vision System	92000.00
Vikram Rao	Sciences	AI Chatbot	88000.00
Simran Kaur	Sciences	Computer Vision System	73000.00
Yash Agarwal	Sciences	Computer Vision System	95000.00
Nandini Iyer	Sciences	AI Chatbot	83000.00
Priya Mehta	Law	Employee Analytics	81000.00
Neha Kapoor	Law	Payroll Automation	69000.00

CORRELATED SUBQUERY

emp_name	salary	dept_id
Aarav Sharma	65000.00	1
Priya Mehta	81000.00	3
Sneha Gupta	68000.00	5
Rahul Joshi	92000.00	2
Karan Malhotra	75000.00	4
Vikram Rao	88000.00	2
Arjun Das	63000.00	1
Kavya Shah	79000.00	3
Nikhil Jain	66000.00	1
Varun Sethi	85000.00	4
Yash Agarwal	95000.00	2
Mohit Bansal	77000.00	4
Aditi Chawla	68000.00	5
Dev Thakur	64000.00	1
Shreya Bose	74000.00	3

EXISTS

dept_id	dept_name
2	Sciences

SIMULATED INTERSECT

emp_id	emp_name
2	Ananya Singh
7	Rahul Joshi
11	Vikram Rao
16	Simran Kaur
21	Yash Agarwal
26	Nandini Iyer

SIMULATED EXCEPT

No data - please run the query to get the data

EXPLAIN - INNER JOIN

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	PRIMARY	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_453k9p6f4.d.dept_id	6	100	null

EXPLAIN - LEFT JOIN

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	d	null	ALL	null	null	null	null	5	100	null
1	SIMPLE	e	null	ref	dept_id	dept_id	5	db_453k9p6f4.d.dept_id	6	100	null